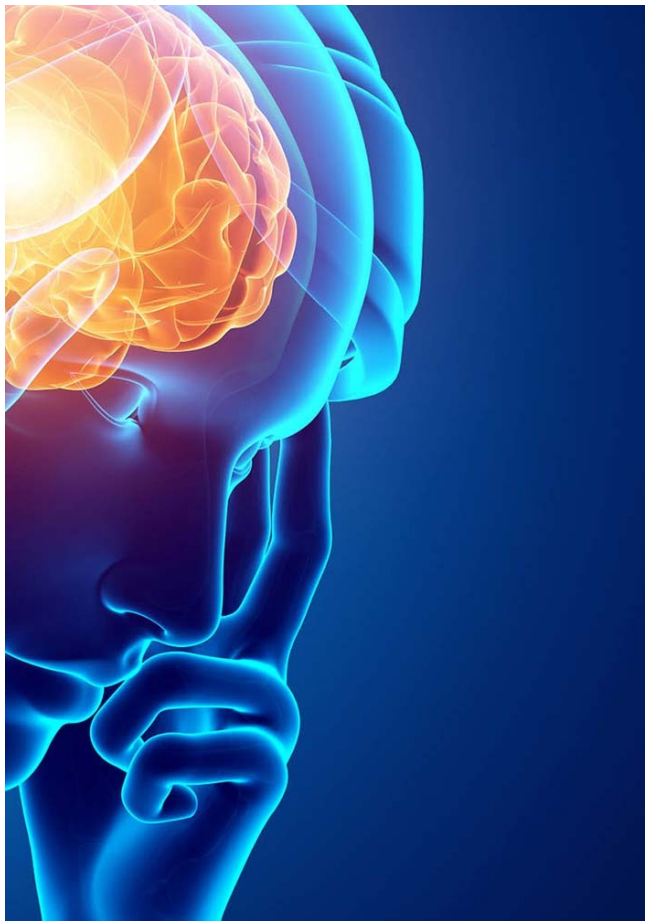
Diseases of the Nervous System

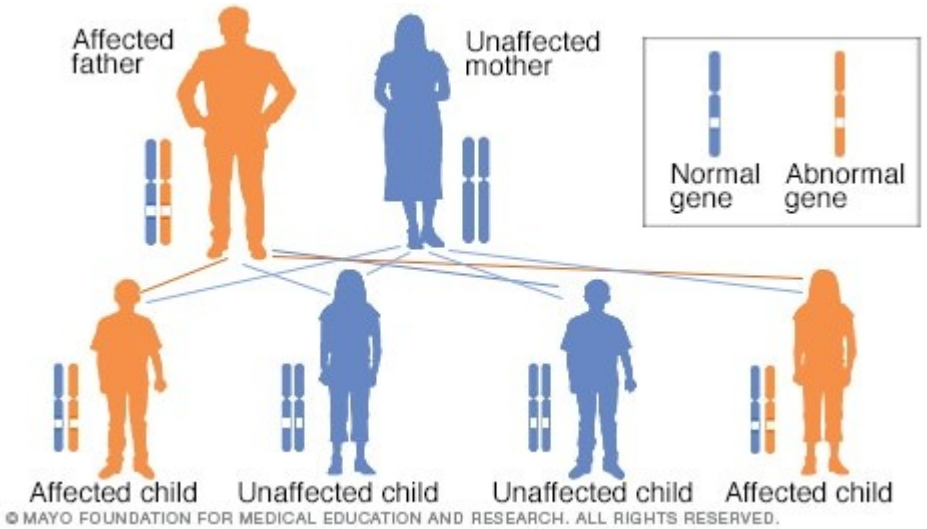


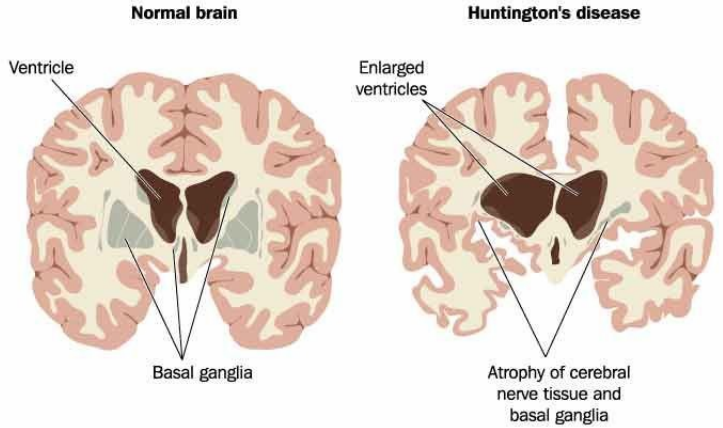
The Nervous System is highly organised

Technology is used to treat diseases

Huntington's Disease

- A single gene disorder – dominant inheritance.
- Abnormal gene:
  - Codes for a mutated protein (huntingtin)
  - Protein damages nerve cells in the brain
  - Large
- Symptoms start after age 40 and are progressive:
  - Physical, mental and emotional changes
  - Progressive dementia, limb flailing.
- Currently incurable, fatal.
- Possible future treatment: Gene therapy

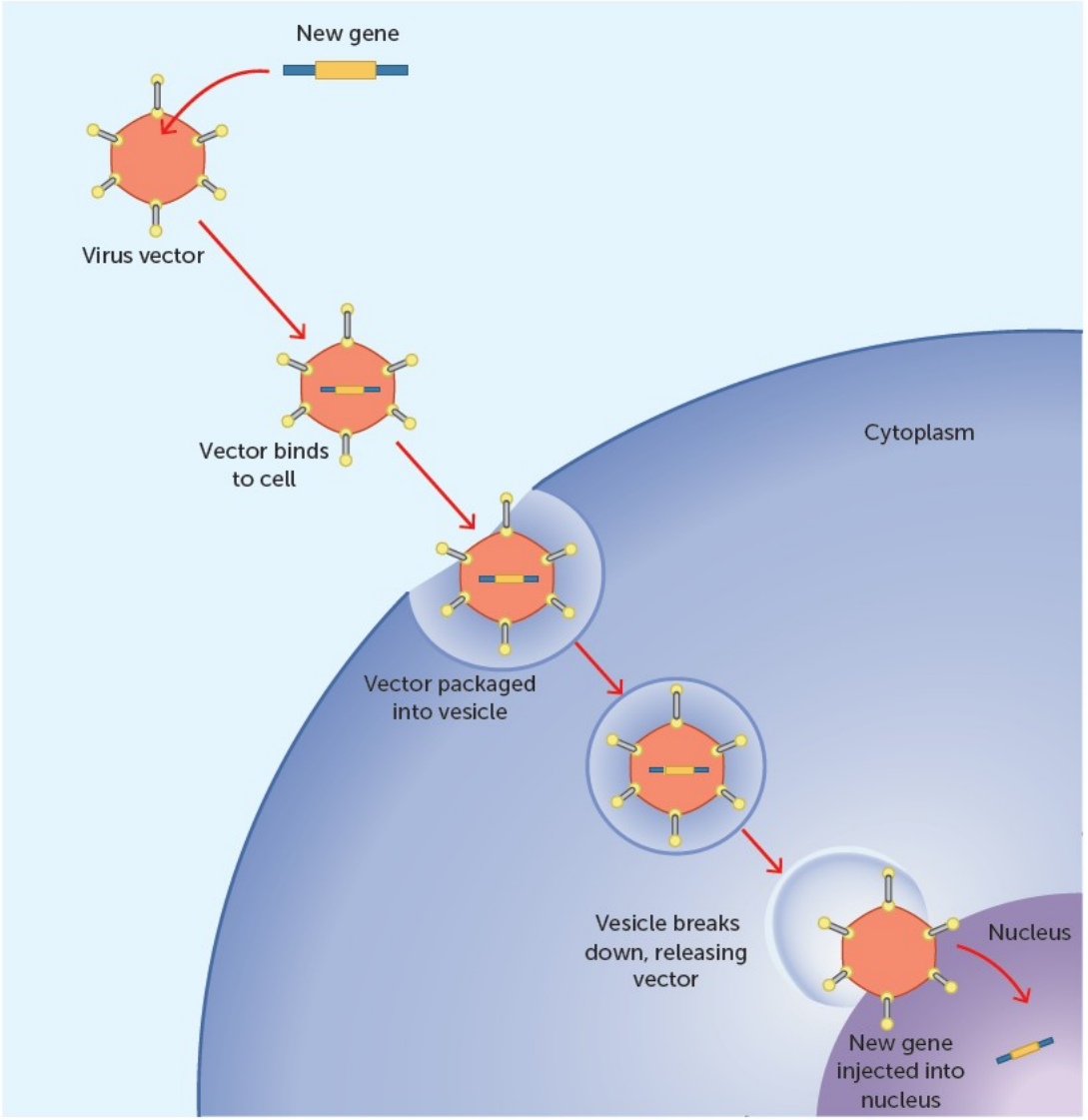




Learning Aim: Describe the cause and disease progression of Huntington's Disease

Gene Therapy

- Aims to treat or cure disease by identifying faulty genes and inserting healthy ones.
- Still in research phase.
- Concept is that a vector (eg a virus)can be used to deliver the desired DNA into the cell, to be incorporated by the nucleus and produce desired protein.
- Areas of possibility:
  - Replacing mutated gene with healthy copy*
    - Eg **Huntington's disease**, Cystic Fibrosis, Type I Diabetes
  - Fixing or inactivating mutant genes*
    - Eg **Huntington's disease**
  - Inserting a new gene that will fight the disease
  - Making the immune system recognise diseased cells.
    - Eg therapy for some cancers

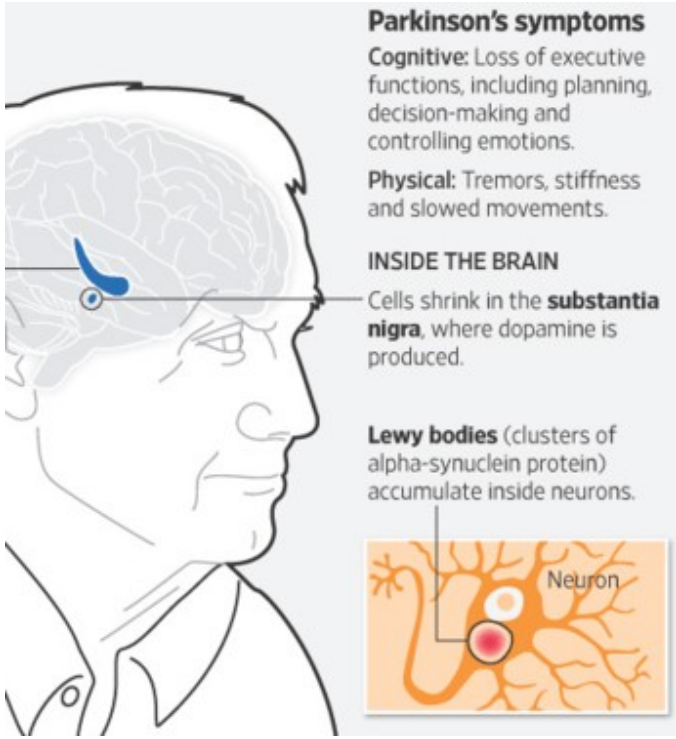


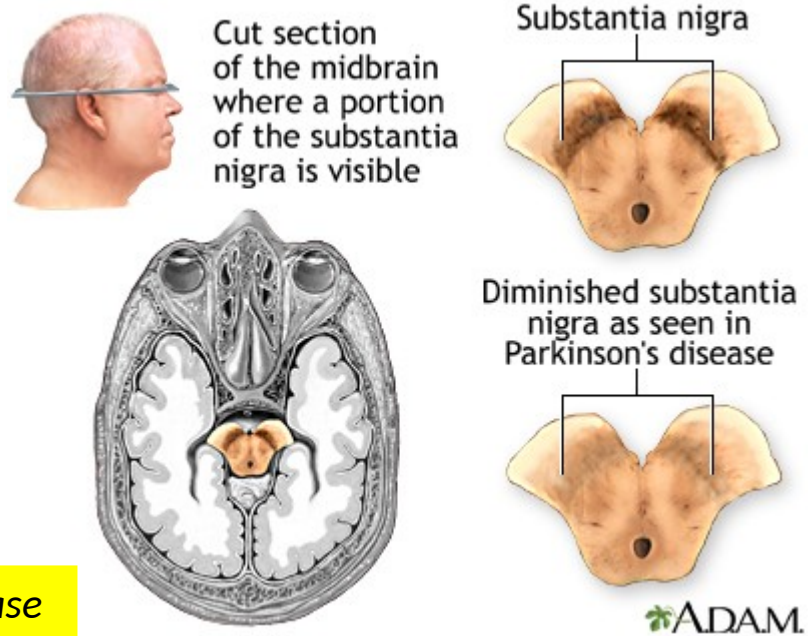
*These ones could be used for Huntington's Disease

Learning Aim: Describe how gene therapy works and why it may be useful in future to treat Huntington's Disease

Parkinson's Disease

- Degenerative (progressive) neurological disease
  - Central NS tissues affected
    - Cell death in substantia nigra (in basal ganglia) decreases dopamine production causing symptoms
    - No identified cause of the cell death
    - Lewy bodies (protein) accumulates inside neurons
  - Symptoms: Impaired motor skills and speech
  - Usually diagnosed later in life (between 60 and 70 years old)
  - No cure, but can be managed.
  - Future cure may be cell replacement therapy

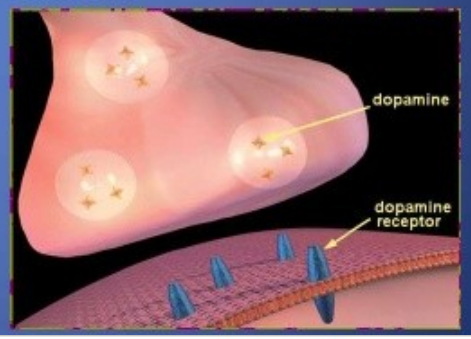




Learning Aim: Describe the disease progression and symptoms of Parkinson's Disease

What is dopamine?

- Dopamine is a neurotransmitter in the brain
- Essential component of the basal ganglia motor loop as well as the neurotransmitter responsible for controlling the exchange of information from one brain area to another
- Plays vital roles in certain behavior such as
  - Movement
  - Cognition
  - Pleasure
  - Motivation



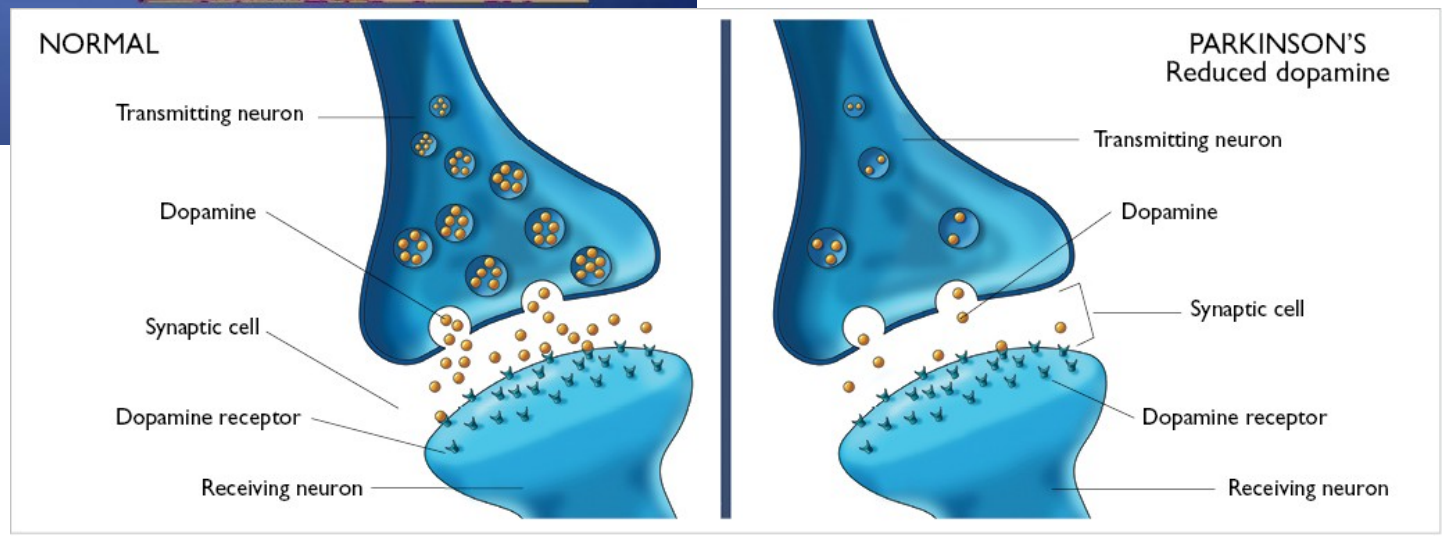
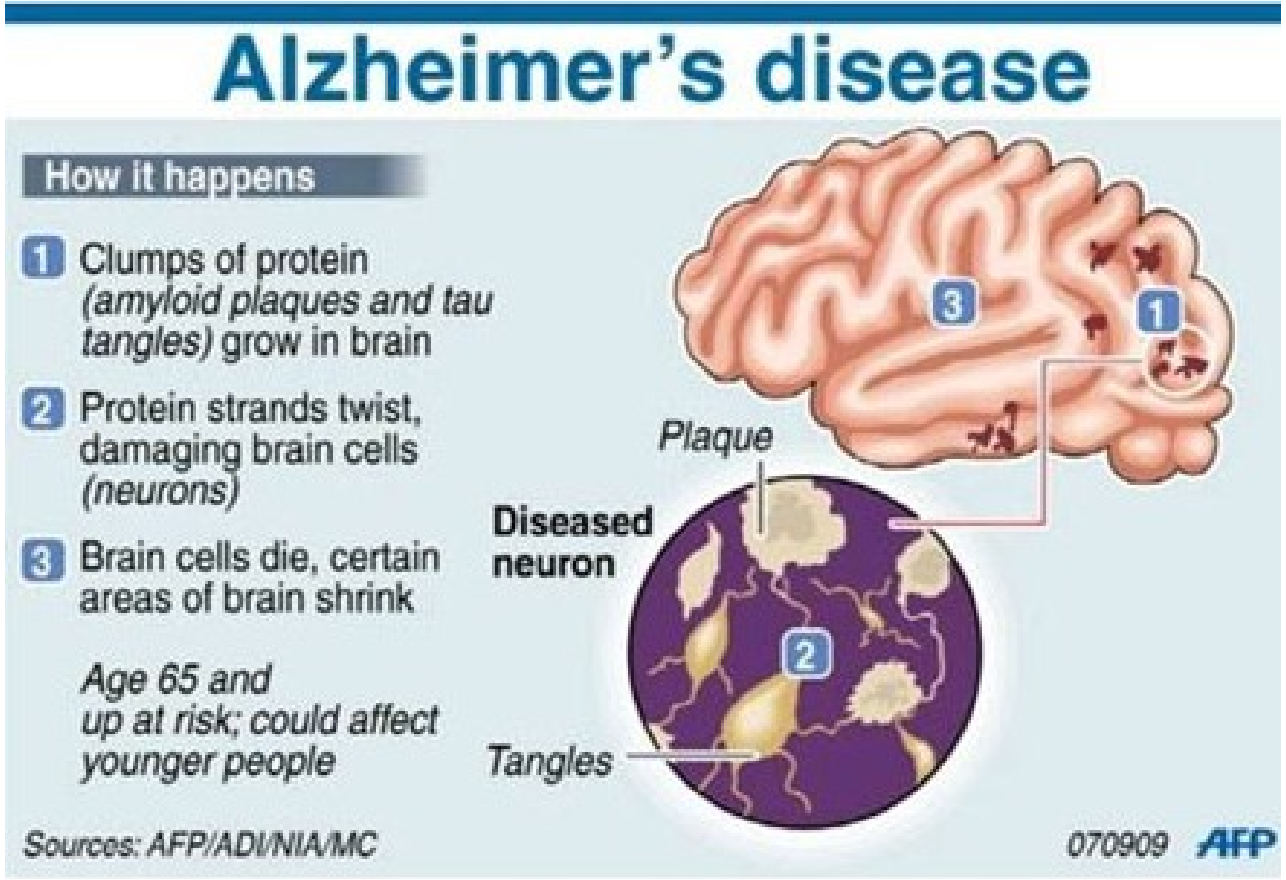


FIGURE 3. Illustration showing low levels of dopamine in a neuron affected by Parkinson's disease (right) and normal levels (left).

Alzheimer's Disease

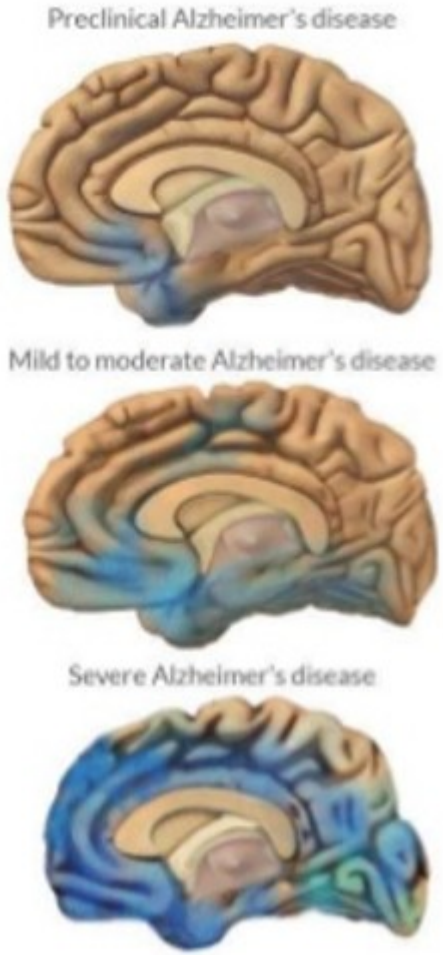
- Progressive Degenerative disease of the brain
- Usually diagnosed after the age of 65
- No clear cause has been established
- Disease progression:

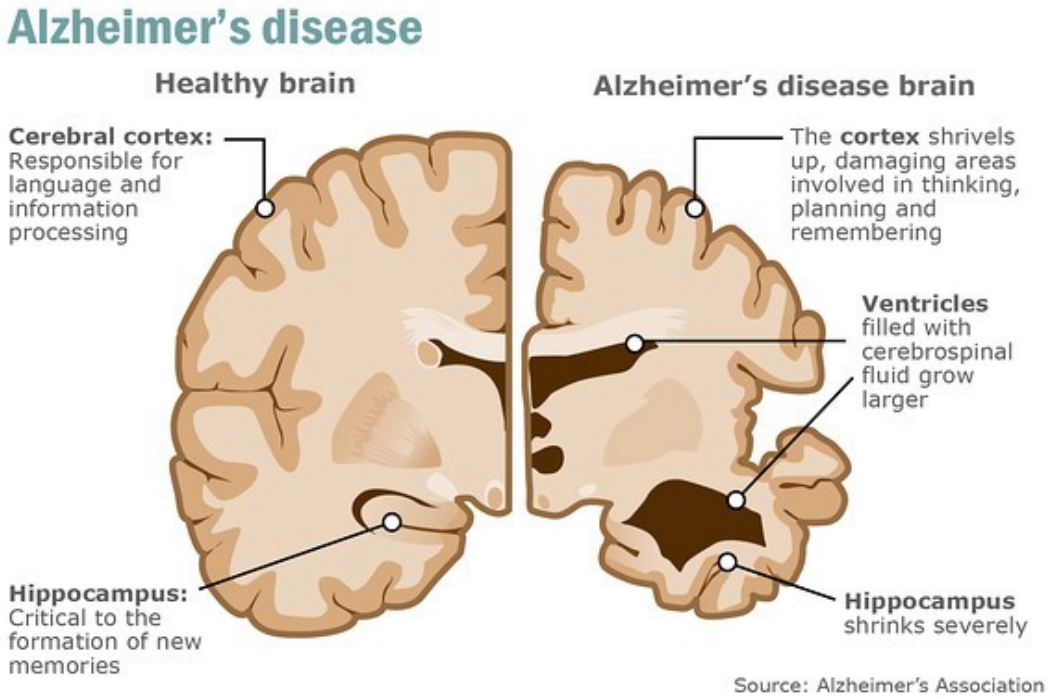


Learning Aim: Describe the disease progression and symptoms of Alzheimer's Disease

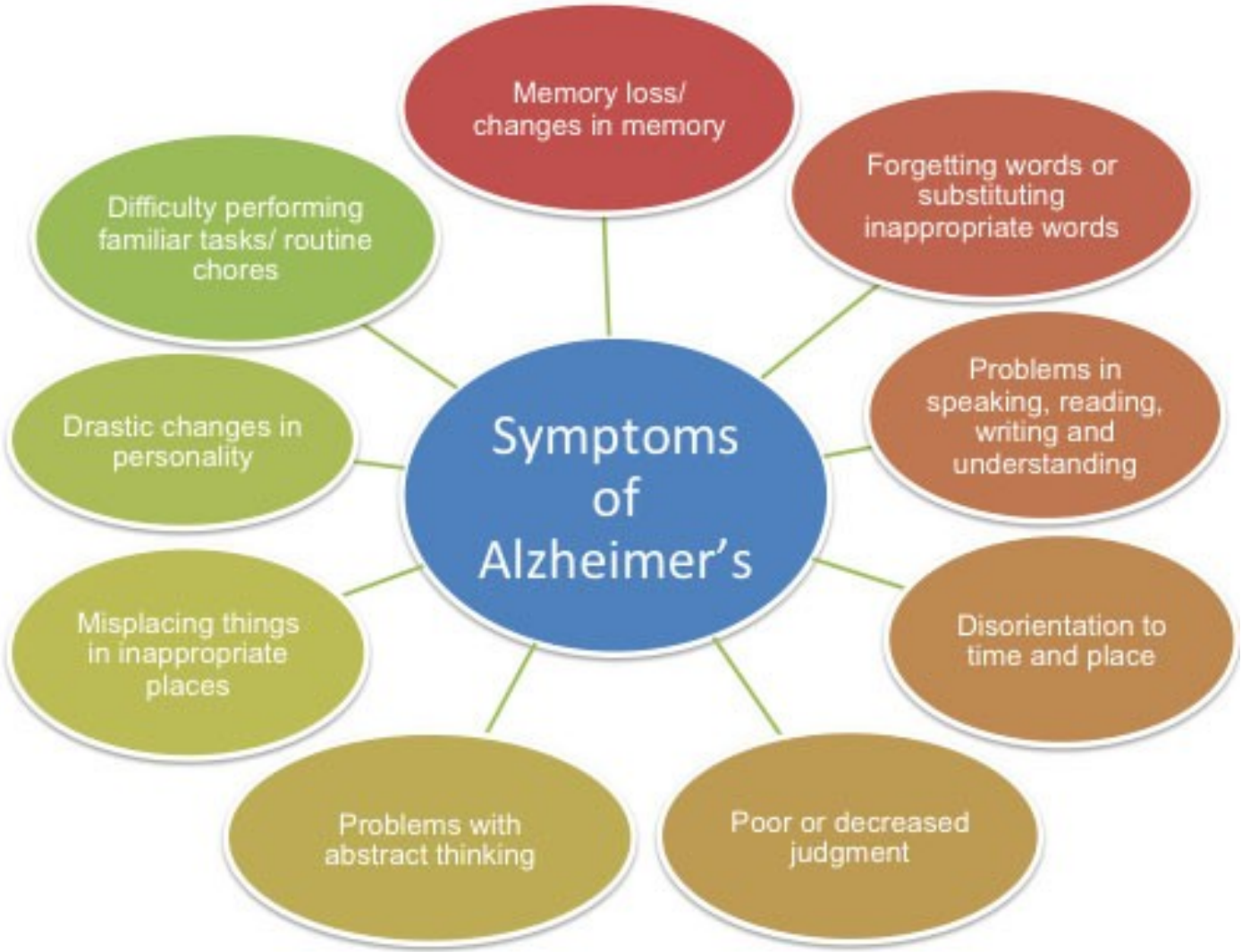
As the disease progresses…

- Neurons lose their ability to function and communicate with each other, and eventually die.
- As it progresses, neurofibrillary tangles spread throughout the brain (shown in blue). Plaques also spread throughout the brain.
- By the final stage, damage is widespread and brain tissue has shrunk significantly.
- Currently no cure – future cure could be cell replacement therapy





Learning Aim: Describe the disease progression and symptoms of Alzheimer's Disease



Symptoms of Alzheimer's disease

Learning Aim: Describe the disease progression and symptoms of Alzheimer's Disease

Cell Replacement Therapy

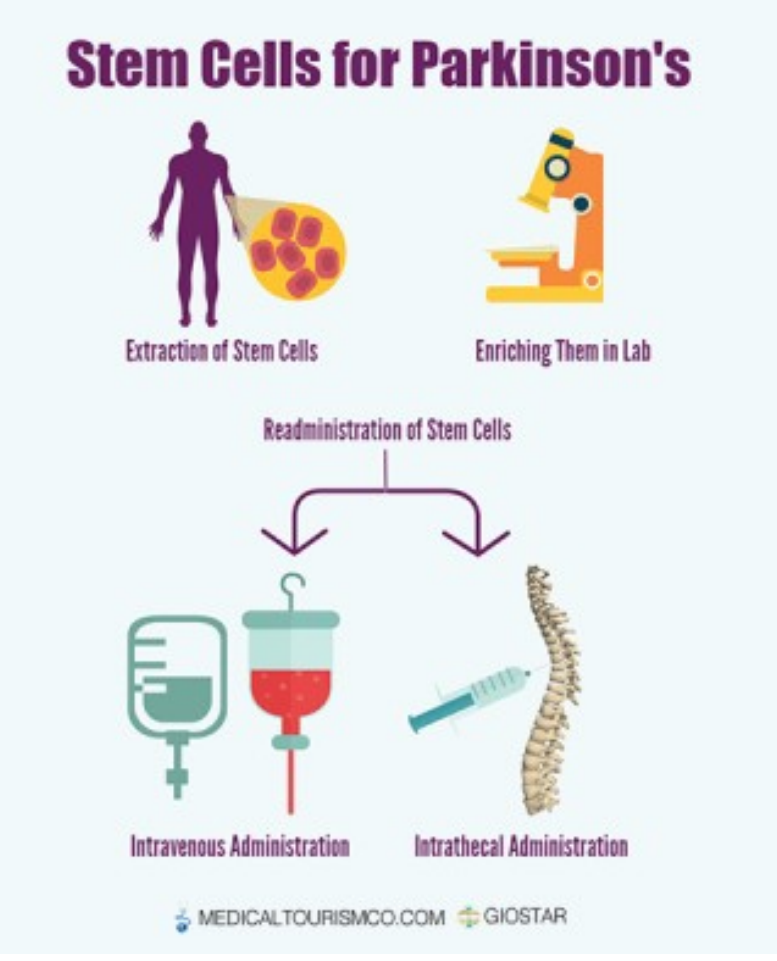
- Could be used in future for neurodegenerative disorders such as Alzheimer's and Parkinson's.

- Replacement of dying neurons with healthy ones derived from stem cells
  - Can grow and differentiate to replace damaged tissue
  - Controversy surrounding use of human embryonic stem cells
  - Other sources of stem cells are being researched

- Technique not yet refined – is a potential future treatment.

Learning Aim: Define cell replacement therapy and discuss how it could be useful in future to treat Parkinson's and Alzheimer's diseases

How Cell Replacement therapy might work once it is through the experimental stage:



- Healthy stem cells are extracted – could be:
  - From the patient – pluripotent, multipotent
  - Embryonic - totipotent
- Stem cells are enriched in the laboratory:
  - Programmed to differentiate into cell type needed
    - Eg dopamine producing neurons for Parkinsons
  - May also be genetically modified to “fix” faulty stem cells
  - Cultured to grow and divide so many are available.
- Stem cells are administered to the patient
  - Grow new, healthy tissue to replace damaged cells
  - Can be administered IV or into spine for Parkinson’s

Learning Aim: Define cell replacement therapy and discuss how it could be useful in future to treat Parkinson’s and Alzheimer’s diseases